Natural Language Processing

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What is NLP?

NLP is the **computational** study of ***language use***, that is, the analysis and design of computational agents that use natural languages to acquire information from other agents, human or machine

Pereira, Fernando C. N., & Grosz, Barbara J. (1993), "Introduction to the Special Volume on Natural Language Processing“, Artificial Intelligence 63: 1-15

Why NLP?

- Learn something about language
- Build something of direct practical use
- As part of a broader program of AI

Bender and ao Vivo, 2020

The Rose Family

By Robert Frost

The rose is a rose,
And was always a rose.
But the theory now goes
That the apple's a rose,
And the pear is, and so's
The plum, I suppose.
The dear only knows
What will next prove a rose.
You, of course, are a rose -
But were always a rose.

https://k16space.blogspot.com/2018/01/poem-analysis-rose-family-by-robert.html

Freepik

Computational Study of Language Use

- Meaning
  - Acquire
  - Represent
  - Retrieve

Marr's Levels of Analysis

David Marr proposes that information processing systems can be understood at three levels:

- Computational level
  - What is the goal?, What are inputs and outputs?
  - What mathematical language can be used to express that goal?
- Algorithmic or representational level
  - What representations and algorithms are used to achieve the computational-level goal?
- Implementation level
  - How is the system implemented, either physically or in software?

Assumptions all the way down

Example

Problem: Modelling people waiting in a line

- Computational level: FIFO queue
- Algorithmic/representational level: Linked list data structure
- Implementation level: memory management, pointers etc.

Krafft and Griffiths, 2018

Processing *a* meaning

- Summarize
- Translate
- Sentiment
- QA
- Generate

Translation

റോബർട്ട് ഫ്രോസ്റ്റിന്റെ ദി
റോസ് ഫാമിലി

റോസ് ഒരു റോസാപ്പൂവാണ്,
എപ്പോഴും ഒരു
റോസാപ്പൂവായിരുന്നു.
എന്നാൽ ഇപ്പോൾ സിദ്ധാന്തം
പോകുന്നു
ആപ്പിൾ ഒരു റോസാപ്പൂവാണ്,
പിയർ ഒരു റോസാപ്പൂവാണ്,
പ്ലം, ഞാൻ കരുതുന്നു.
പ്രിയയ്ക്ക് മാത്രമേ അറിയൂ
അടുത്തതായി ഒരു റോസാപ്പൂവ്
എന്താണെന്ന് തെളിയിക്കും.
തീർച്ചയായും, നീ ഒരു
റോസാപ്പൂവാണ് -
പക്ഷേ എപ്പോഴും ഒരു
റോസാപ്പൂവായിരുന്നു.

रॉबर्ट फ्रॉस्ट की कविता "द रोज़
फैमिली"

गुलाब तो गुलाब है,
और हमेशा से गुलाब ही रहा है।

लेकिन अब यह माना जाता है,
कि सेब भी गुलाब है,
और नाशपाती भी, और
शायद बेर भी।

भगवान ही जाने,
कि आगे कौन गुलाब साबित होगा।

तुम तो बेशक गुलाब हो -
लेकिन हमेशा से गुलाब ही रहे हो।

ராபர்ட் ஃப்ரோஸ்டின் ரோஜா
குடும்பம்

ரோஜா ஒரு ரோஜா,
எப்போதும் ஒரு ரோஜாவாகவே
இருந்தது.
ஆனால் இப்போது கோட்பாடு
செல்கிறது
ஆப்பிள் ஒரு ரோஜா,
பேரிக்காய் ஒரு ரோஜா,
பிளம், நான் நினைக்கிறேன்.
அன்பானவருக்கு மட்டுமே
தெரியும்
அடுத்து ரோஜாவை எது
நிரூபிக்கும்.
நீங்கள் நிச்சயமாக ஒரு ரோஜா -
ஆனால் எப்போதும் ஒரு
ரோஜாவாகவே இருந்தீர்கள்.

Key Applications and Tasks

- Applications
  - Machine translation
  - Speech recognition
  - Search engines
  - Spell and Grammar checkers
  - Summarization
  - Question Answering
  - Natural language generation
  - Chatbots
  - Sentiment analysis
  - ….
- Tasks
  - Part-of-speech tagging
  - Dependency parsing
  - Named-Entity Recognition
  - Language models
  - Semantic role labeling
  - Co-reference resolution
  - Event extraction
  - …

We have ChatGPT. Why should we even study NLP?

Why NLP is hard?

- Optimization
- Space and time
- Cognition
- Frames
- Subjectivity (politics, religion, culture, poetry)
- Context and intent
- The world

Easy things are difficult: Moravec's paradox
It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility

M. Minsky, 1987
Moravec, 1988

Form/Meaning/Communicative Intent

John: "Mary! The telephone!"

Mary: “I am in the bath."

Yes, John, I heard you and I know that when you said "Mary! The telephone!" you meant that I should answer it because otherwise you would have answered it yourself instead of calling for me, but I don't want to answer it because I am having a bath and, as you know, when people are having a bath they do not like to get out of the tub and go into the hall to answer the telephone because they are wet and soapy and get cold and produce puddles on the floor and the bathwater cools off...

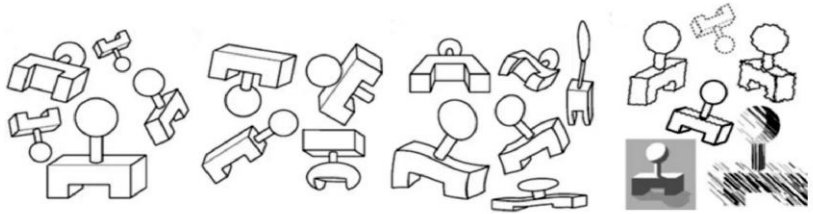
NE Enkvist, 1988

Gestalt Principles

Gestalt Rule: **Emergence**

Interaction Design Foundation
interaction-design.org

Gestalt Rule: **Invariance**



Interaction Design Foundation
interaction-design.org

Gestalt Principles

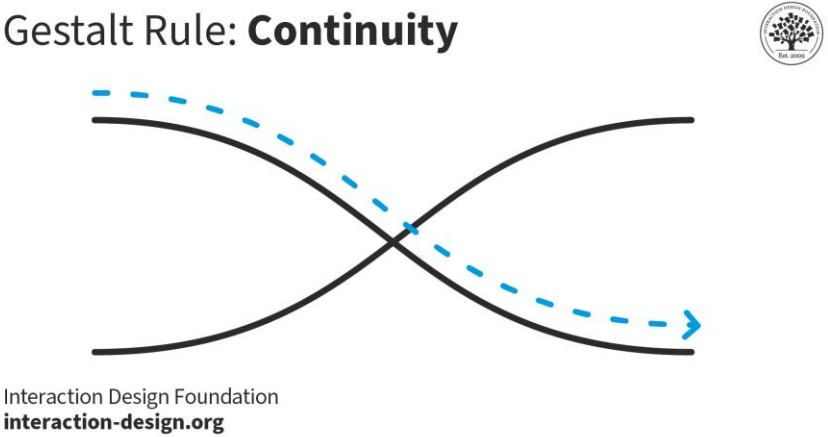
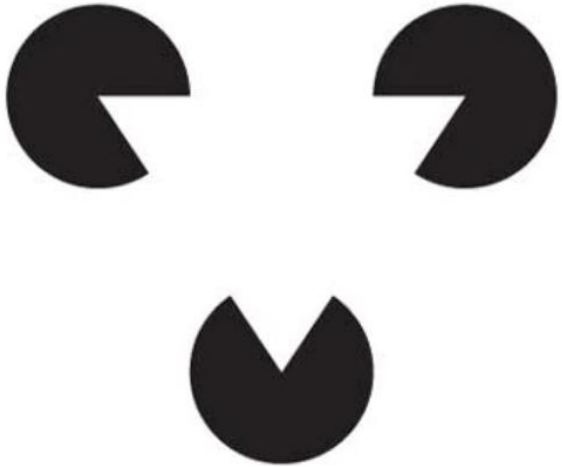


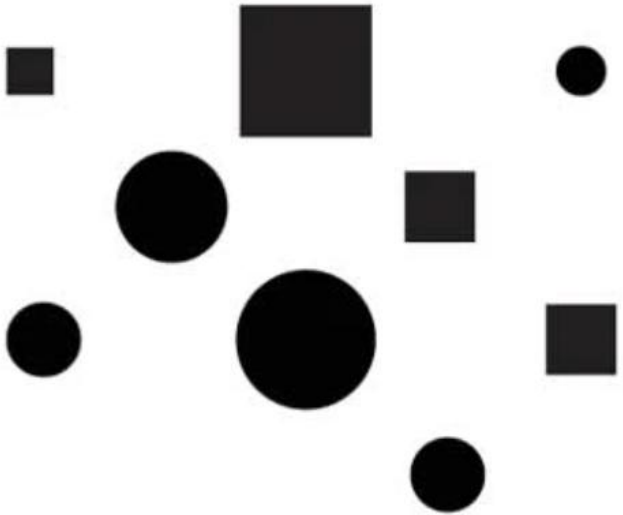
Figure and Ground

https://eportfolios.capilanou.ca/yunhu/2020/09/29/gestalt-principles-example-figure-ground/

Gestalt Principles



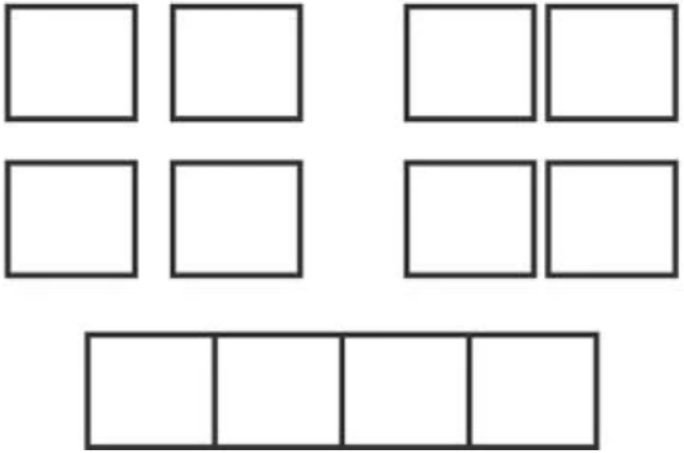
Subjective contour



Similar objects are visually grouped

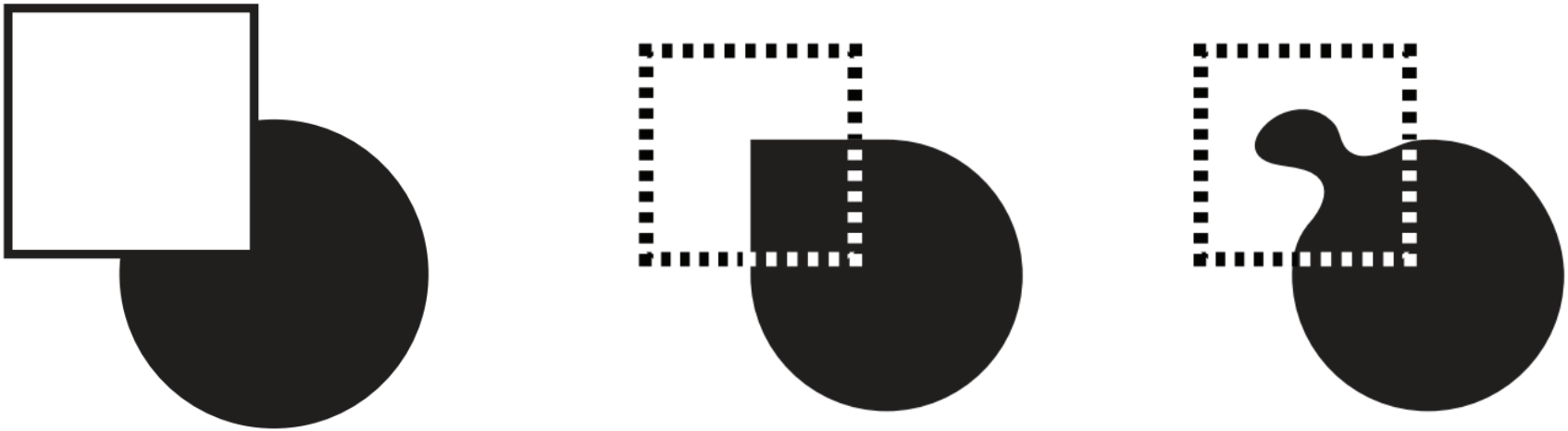
https://www.nature.com/articles/nmeth1110-863

Gestalt Principles



https://www.nature.com/articles/nmeth1110-863

Hidden Shapes

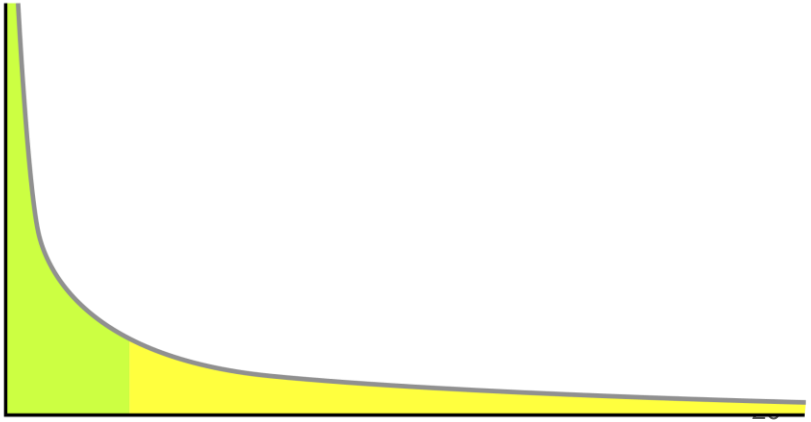


https://www.nature.com/articles/nmeth1210-941.pdf

NLP is hard

But, we have some known patterns or conventions:

- Grammar and syntax
- Selectional constraints or selectional preferences
  - John **ate** a sandwich
  - John **ate** my whole day
  - John **ate** my pen
- Power law



https://en.wikipedia.org/wiki/Power_law

Text or utterance is NOT data

Language and People

The common misconception is that language has to do with **words** and what they mean.

It doesn't.

It has to do with **people** and what ***they*** mean.

Decisions we make about our data, methods, and tools are tied up with their impact on people and societies.

Herbert H. Clark & Michael F. Schober, 1992
Yulia Tsvetkov, 2020

Course logistics

- Evaluation
  - Mid-semester examination (15%): **Saturday 07 March 2026: 09.00 am**
  - End-semester examination (30%)
  - Reading assignments (25%): Weekly/Bi-weekly
  - Project (30%)

- TAs
  - Abigairl Nyasha Chigwededza
  - John Salvin
  - Jotipriya Das
  - Yedavali Siva Kumar

Reading Assignments

- Read uploaded paper(s) for a week
- Write maximum 2 pages ***about*** the paper(s)
  - What is *interesting* in the paper(s)? Why?

  (It is unlikely that you may find nothing interesting)
  - What are possible *implications* of the paper(s)?
- Critical thinking: https://www.youtube.com/watch?v=zDQTx1k2TLo
- Use of AI tools
- Turnitin-based submission

NLP Projects

Team

- 3 members
- all the members should not be of the same gender
- at least one B. Tech student and one PG student

NLP Applications

- Evaluating dialogue systems/LLMs
- Discourse: Coreference resolution
- Ethics, Bias, and Fairness
- Generation
- Information Extraction: named entities, events, temporal expressions
- Topic models
- Interpretability and Analysis of Models for NLP
- Machine Translation
- Question Answering
- Sentiment Analysis
- Summarization

https://2025.aclweb.org/calls/main_conference_papers/

Coreference resolution

(1) John$_{1,1}$ was very happy yesterday.
(2) So, he$_{2,1}$ called up Mike$_{2,2}$.
(3) He$_{3,1}$ wanted to meet him$_{3,2}$ urgently.

Project proposal

- Motivation
  - which NLP task/application do you plan to do;
  - which aspect of the problem / task did your group plan to work on (accuracy, interpretability, etc.);
  - provide some reasons for your choice
- Approach
  - Describe the algorithms and machine learning models you plan to use in your project.
  - Using equations is not necessary but if you do, use a clear mathematical style to explain your model(s).
- Data
  - Exactly which data set do you plan to use
  - What is the evaluation measure for the data set and what is the baseline

https://anoopsarkar.github.io/advanced-nlp-class/project.html

Deliverables

- Dataset
  - **Who annotated the data?**
  - Annotation guidelines
  - Annotation interface
  - Create a small dataset for one Indian language
- GitHub repository
- Project report
- Presentation/Poster

Evaluation-I

- Project (30%):
  - Problem statement--- Deadline: 30-Jan
  - Code
  - Project report
  - Presentation/Poster
  - Data Annotation

Evaluation-II

Project Report

- **Motivation**: Which aspect of the problem / task did your group choose to improve and reasons for your choice.
- **Approach:** Describe the algorithms and machine learning models used in your project. Use a clear mathematical style to explain your model(s).
- **Data:** Exactly which data files were used; also include here any external data that was not provided to you.
- **Code:** Provide exactly which code was used in your project not written by your group (e.g. use of an aligner from an open-source project).
- **LLMs:** exact details of LLMs used and prompts used for evaluation
- **Experimental Setup:** Describe what kind of evaluation you are doing and which methods you are comparing against each other.
- **Results:** Include a detailed comparison of different methods.
- **Analysis of the Results:** Did you improve over the baseline. Why or why not?
- **Future Work:** What could be fixed in your approach. What you did not have time to finish, but you think would be a useful addition to your project.

https://anoopsarkar.github.io/advanced-nlp-class/project.html

Evaluation-III

- Presentation/Poster
  - Goal is clear?
  - Implementation is clear?
  - Model was clear?
  - Did provided example make things clear?
  - Experimental evaluation was clear?
  - Adequate comparison to previous work?
  - Poster layout and oral explanation.
  - Analysis of the output.
  - Overall quality.

https://anoopsarkar.github.io/advanced-nlp-class/project.html

Evaluation-IV

Code: GitHub repository

- README file
  - Configuration, Installation and Operating instructions
  - A file manifest (a list of files in the directory or archive)
  - Copyright and licensing information
  - Contact information for the distributor or author
  - A list of known bugs and/or issues
  - Troubleshooting instructions
  - Credits and acknowledgments
- Link to datasets and other relevant repositories

We will clone your repository and follow the instructions in the README.
The code MUST reproduce the results reported in the project report.

https://en.wikipedia.org/wiki/README

Final Project Evaluation

- Originality
- Substance (amount of work done for the project)
- Well documented use of prior results from research papers
- Clarity of the writing
- Quality of experimental design
- Quality of evaluation and results
- Theoretical insights / Practical insights
- Group work (did the group work effectively together)
- Overall score (based on the above criteria, but can include other factors like overall polish or creativity)

https://anoopsarkar.github.io/advanced-nlp-class/project.html

NLP on the web

- Stanford https://web.stanford.edu/class/cs224n/
- CMU https://www.phontron.com/class/anlp-fall2024/
- UW https://msclar.github.io/cse447-au24/
- NPTEL https://onlinecourses.nptel.ac.in/noc23_cs45/preview
- Jay Alammar https://jalammar.github.io/
- 3Blue1Brown https://www.3blue1brown.com/
- …

Resources

- Jacob Eisenstein, 2019. Introduction to Natural Language Processing.
- Daniel Jurafsky and James Martin, 2021. Speech and Language Processing
- Emily M. Bender, 2013. Linguistic Fundamentals for Natural Language Processing: 100 Essentials from Morphology and Syntax
- Emily M.Bender and Alex Lascarides, 2019. Linguistic Fundamentals for Natural Language Processing II: 100 Essentials from Semantics and Pragmatics.
- Yoav Goldberg, 2017. Neural Network Methods in Natural Language Processing.

Resources

- Linguistics Olympiad
  - https://ioling.org/
  - https://naclo.org/
  - https://ltrc.iiit.ac.in/plo/

Important Videos

- Steven Pinker: Linguistics as a Window to Understanding the Brain
  - https://www.youtube.com/watch?v=Q-B_ONJIEcE
- Natural Language Processing by Prof Sutanu Chakraborti
  - https://www.youtube.com/watch?v=lFgV8ajJjsc

Something extra

- *Story of Your Life* by Ted Chiang

or the movie: *Arrival*

- *Babel-17* by Samuel R. Delany
- *Flatland* by Edwin A. Abbott
- *The Embedding* by Ian Watson
- Explore Scifi stories based on Linguistics